

1. Give the order of the differential equation that has $y^3 = Cx^4 - 3x$ as its general solution and find the equation:
- (a) 1; $y' = \frac{y^3 + 3x}{3xy^2}$
 - (b) 1; $y' = \frac{4y^3 + 9x}{3xy^2}$
 - (c) 2; $y' = \frac{4y^3 + 12x}{3xy^2}$
 - (d) 0; $y' = \frac{4y^3 + 12}{3xy^2}$
 - (e) None of the above.
2. Give the order of the differential equation that has $y = C_1 + C_2x^3 + 2x^2$ as its general solution and find the equation.
- (a) 1; $x^2y'' - 2y = x$.
 - (b) 2; $xy'' - 2y' - 4x = 0$.
 - (c) 3; $y'' - 2y = 4x$.
 - (d) 2; $xy'' - 2y' = -4x$.
 - (e) none of the above.
3. The general solution of $x^2y' = 4x^4e^{2x} + 2xy$ is:
- (a) $y = 2x^2e^{2x} + Cx^2$.
 - (b) $y = 2x^2e^{2x} + C$.
 - (c) $y = 2e^{2x} + Cx^2$.
 - (d) $y = 2x^2e^{2x} - xe^{2x} + Cx^2$.
 - (e) none of the above.
4. The general solution of $(x^2y + 5y)y' = 2xy^2 + 6x$ is:
- (a) $y^2 + 3 = (x^2 + 5) + C$
 - (b) $y^2 = C(x^2 + 5)^2 - 3$
 - (c) $y^2 + 3 = (x^2 + 5)^2 + C$
 - (d) $y^2 = C(x^2 + 5) - 3$
 - (e) None of the above

5. The general solution of $xy' = x^3y^3 \sin 2x - x^2y$ is:

- (a) $y^2 = \frac{1}{x^2 \cos 2x + Cx^2}$
- (b) $y = \frac{1}{\sqrt{x^2 \cos 2x + C}}$
- (c) $y^2 = \frac{1}{C - x^2 \cos 2x}$
- (d) $y = \sqrt{x^2 \sin 2x + Cx^2}$
- (e) none of the above.

6. The general solution of $xyy' = xye^{y/x} + y^2$ is:

- (a) $y = -x \ln(C + \ln x)$
- (b) $y = x \ln(\ln x - C)$
- (c) $y = x \ln(C/x)$
- (d) $y = -x \ln(C - \ln x)$
- (e) none of the above.

7. The general solution of $x^2y y' = x^3 + 2xy^2$ is:

- (a) $y^2 = Cx^4 - x^2$
- (b) $y^2 = x^4 - Cx^2$
- (c) $y^2 = x^2 \ln x^2 + Cx^2$
- (d) $y^2 = x^2 \ln Cx^2$
- (e) None of the above.

8. The family of orthogonal trajectories of $y^3 = Cx^4 - 2$ is:

- (a) $3x^2 + 4y^2 - \frac{8}{y} = C$
- (b) $x^2 + 2y^2 + 8 \ln y = C$
- (c) $3x^2 + 4y^2 - \frac{16}{y} = C$
- (d) $4y^2 + \frac{16}{y} = 3x^2 + C$
- (e) none of the above.

9. A sample of 100 grams of radioactive material was present initially and after 3 hours the sample lost 20% of its mass. Find the amount of material that remains after 9 hours and give the half-life of the material?

- (a) $A(9) = 100 \left(\frac{4}{5}\right)^3$; $T = \frac{-3 \ln 2}{\ln(4/5)}$
- (b) $A(9) = 100 \left(\frac{3}{5}\right)^2$; $T = \frac{-2 \ln 2}{\ln(3/5)}$
- (c) $A(9) = 100 \left(\frac{4}{5}\right)^9$; $T = \frac{-9 \ln 2}{\ln(4/5)}$
- (d) $A(9) = 100 \left(\frac{3}{5}\right)^{-3}$; $T = \frac{-3 \ln 2}{\ln(3/5)}$
- (e) None of the above.

10. A flotilla of sea turtles obeys the population growth law. Initially the flotilla had 75 turtles. Ten years later the flotilla had 100 turtles. Find the number of sea turtles at any time t and give the doubling time of the flotilla?

- (a) $A(t) = 75 \left(\frac{3}{4}\right)^{t/10}$; $T = \frac{10 \ln 2}{\ln(3/4)}$
- (b) $A(t) = 100 \left(\frac{4}{3}\right)^{10t}$; $T = \frac{\ln 2}{\ln(4/3)}$
- (c) $A(t) = 75 \left(\frac{4}{3}\right)^{10t}$; $T = \frac{\ln 2}{10 \ln(4/3)}$
- (d) $A(t) = 75 \left(\frac{4}{3}\right)^{t/10}$; $T = \frac{10 \ln 2}{\ln(4/3)}$
- (e) None of the above.

11. A disease is spreading through a small cruise ship with 200 passengers. Let $P(t)$ be the number of people who have disease at time t . The disease is spreading at a rate proportional to the product of the time elapsed and the number of people that are not sick. Suppose that 20 people have the disease initially. The mathematical model for the spread of the disease is:

- (a) $\frac{dP}{dt} = k(200 - P)$, $P(0) = 20$.
- (b) $\frac{dP}{dt} = kt(200 - P)$, $P(0) = 20$.
- (c) $\frac{dP}{dt} = ktP$, $P(0) = 20$.
- (d) $\frac{dP}{dt} = kP(200 - P)$, $P(0) = 20$.
- (e) None of the above.

12. Refer to Problem 11. Suppose that 50 people are sick after 4 days. Then the number of people that are sick at any time t is given by:

- (a) $P(t) = 200 - 180 \left(\frac{5}{6}\right)^{t/4}$

- (b) $P(t) = 20 \left(\frac{5}{2}\right)^{t^2/16}$
- (c) $P(t) = 200 - 180 \left(\frac{5}{6}\right)^{t^2/16}$
- (d) $P(t) = \frac{200}{2 + 18(5/6)^{t/4}}$
- (e) None of the above.
13. $u_1 = 2x^{-3}$, $u_2 = 3x^4 + 5x^{-3}$, $u_3 = -2x^4 - x^{-3}$ are solutions of a second order, linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$. The general solution of the equation is:
- (a) $y = C_1(2x^{-3}) + C_2(3x^4) - C_3(2x^4)$
- (b) $y = C_1x^{-3} + C_2x^4 + C_3x^4$
- (c) $y = C_1x^3 + C_2x^{-4}$
- (d) $y = C_1x^{-3} + C_2x^4$
- (e) None of the above.
14. $y'' - \frac{2}{x}y' - \frac{10}{x^2}y = 0$ has solutions of the form $y = x^r$. Find a solution basis for the equation and give their Wronskian.
- (a) $\{y_1 = x^{-2}, y_2 = x^5\}$, $W(y_1, y_2) = 3x^2$
- (b) $\{y_1 = x^2, y_2 = x^{-5}\}$, $W(y_1, y_2) = -7x^{-2}$
- (c) $\{y_1 = x^{-2}, y_2 = x^5\}$, $W(y_1, y_2) = 7x^2$
- (d) $\{y_1 = x^2, y_2 = x^{-5}\}$, $W(y_1, y_2) = -3x^{-2}$
- (e) None of the above.
15. Find the solution of the initial-value problem $x^2 y'' - 6y = 0$, $y(1) = 6$, $y'(1) = -2$. (Hint: The equation has solutions of the form $y = x^r$.)
- (a) $y = 2x^3 + 4x^{-2}$
- (b) $y = 4x^6 - 2x^{-1}$
- (c) $y = 4x^3 + 2x^{-2}$
- (d) $y = 2x^{-6} + 4x$
- (e) None of the above.

16. The general solution of $y'' - 6y' + 34y = 0$ is:

- (a) $y = C_1 e^{5x} \cos 3x + C_2 e^{5x} \sin 3x$
- (b) $y = C_1 e^{2x} + C_2 e^{17x}$
- (c) $y = C_1 e^{-2x} + C_2 e^{17x}$
- (d) $y = C_1 e^{3x} \cos 5x + C_2 e^{3x} \sin 5x$
- (e) None of the above.

17. A fundamental set of solutions of $y'' - 4y' - 12y = 0$ is:

- (a) $\{e^{4x}, e^{-3x}\}$
- (b) $\{e^{6x}, e^{-2x}\}$
- (c) $\{e^{2x}, e^{-6x}\}$
- (d) $\{e^{-4x}, e^{3x}\}$
- (e) None of the above.

18. The general solution of $y'' + 10y' + 25y = 0$ is:

- (a) $y = C_1 e^{5x} + C_2 x e^{5x}$
- (b) $y = C_1 e^{-5x} + C_2 e^{5x}$
- (c) $y = C_1 e^{5x} + C_2 x e^{-5x}$
- (d) $y = C_1 e^{-5x} + C_2 x e^{-5x}$
- (e) None of the above.

19. A solution basis for $y'' + 12y' + 36y = 0$ is:

- (a) $\{e^{6x}, x e^{6x}\}$
- (b) $\{e^{6x}, e^{-6x}\}$
- (c) $\{e^{-6x}, x e^{-6x}\}$
- (d) $\{e^{4x}, x e^{9x}\}$
- (e) None of the above.

20. The second order linear homogeneous differential equation that has $y = 2e^{-2x} - e^{4x}$ as a solution is:

- (a) $y'' - 2y' - 8y = 0$

- (b) $y'' - 6y' + 8y = 0$
- (c) $y'' + 2y' - 8y = 0$
- (d) $y'' - 6y' - 8y = 0$
- (e) None of the above.

21. The second order linear homogeneous differential equation that has $y = 4xe^{-3x}$ as a solution is:

- (a) $y'' - 6y' + 9y = 0$
- (b) $y'' + 3y' = 0$
- (c) $y'' + 6y' + 9y = 0$
- (d) $y'' - 9y = 0$
- (e) None of the above.

22. The second order linear homogeneous differential equation that has $y = 3e^{-2x} \cos 2x$ as a solution is:

- (a) $y'' - 4y' + 12y = 0$
- (b) $y'' + 4y' + 8y = 0$
- (c) $y'' - 8y' + 8y = 0$
- (d) $y'' - 4y' + 8y = 0$
- (e) None of the above.

23. The second order linear differential equation that has

$$y = 3e^{2x} - 2xe^{-3x}$$

as a solution is:

- (a) $y'' + y' - 6y = 0$
- (b) $y'' - y' - 6y = 0$
- (c) $y'' - 5y' + 6y = 0$
- (d) $y'' + 5y' + 6y = 0$
- (e) None of the above.